

TD - Practice

Validating “identities”

For each of these “identities,” determine whether it is *valid* or *invalid* for all θ .

If valid, derive it from more basic sine-cosine identities.

Problems

1. $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$
2. $\cos(2\theta) = 1 - 2\sin^2 \theta$
3. $\cos(2\theta) = 2\cos^2 \theta + 1$
4. $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$
5. $\tan(2\theta) = \frac{2 \tan \theta}{1 + \tan^2 \theta}$
6. $\tan^2 \theta = \sec^2 \theta - 1$
7. $\csc^2 \theta = 1 - \cot^2 \theta$
8. $\sin^2 \theta = 1 + \cos^2 \theta$
9. $\sin(2\theta) = 2 \sin \theta \cos \theta$
10. $\sin^2 \theta = \frac{1 + \cos(2\theta)}{2}$
11. $\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$
12. $\cos^2 \theta = \frac{1 - \cos(2\theta)}{2}$
13. $\tan(\pi + \theta) = \tan \theta$
14. $\tan(\pi - \theta) = \tan \theta$
15. $\csc(-\theta) = -\csc \theta$
16. $\sec(-\theta) = -\sec \theta$
17. $\cot^2 \theta + 1 = \csc^2 \theta$
18. $1 + \cot^2 \theta = \sec^2 \theta$